Abdullah Bin Sharif

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EDUCATION

Texas Tech University, Lubbock, Texas — Graduation Spring 2028

Bachelor of Science in Computer Engineering, Minor in Mathematics — **GPA: 3.61**

Relevant Coursework- Robotics Project Lab, Electronics, Microcontroller with Assembly, Modern Digital System Design, Microprocessor Architecture (in progress), Data Structures (in progress)

EXPERIENCE

IT Enterprise Endpoints, Texas Tech University

IT Student Supervisor (Tech III) — April 2026 – Present

IT Student Technician (Tech II) — January 2026 – March 2026

IT Student Analyst (Tech I) — November 2024 – December 2025

- Supervise and work alongside a team of 9 student technicians, leading appointments, walk-up support, and hands-on repairs across residence halls.
- Serve as direct point of contact to the Residence Hall Support manager, reporting incidents, coordinating office needs, and resolving operational issues.
- Troubleshoot Windows, macOS, hardware, software, and network issues for Texas Tech residence hall students.
- Configure network access, printers, peripherals, software installations, system updates, and malware remediation.
- Collaborate with campus IT teams to diagnose recurring technical issues and improve service reliability for over 8,000 campus residents.

SKILLS

Programming: Python, MicroPython, C, C++, MATLAB, Verilog, MIPS Assembly, AVR Assembly, HTML

Tools: KiCad, LTspice, MPLAB, Arduino, Raspberry Pi, IoT, NumPy, Git/GitHub

PROJECTS

Autonomous Soccer Rover ***(Raspberry Pi Pico, Embedded Systems, PCB Design, Motor Control)***

Movement System Lead — January 2026 – May 2026

- Designed and tested a custom discrete MOSFET-based H-Bridge motor driver for bidirectional DC motor control, later transitioning to an L298N-based architecture to improve rover maneuverability and current handling under load conditions.
- Developed and optimized a custom PCB in KiCad, resolving grounding and trace-routing issues to improve electrical stability, reduce noise, and enhance overall system reliability.
- Implemented an overcurrent protection subsystem using current-sense resistors, comparator-based protection circuitry, and LM324AN op-amp signal amplification to protect drivetrain electronics during high-load operation.
- Programmed autonomous movement and navigation logic in MicroPython on a Raspberry Pi Pico 2, integrating differential motor control, ultrasonic sensing, obstacle avoidance, and directional navigation capabilities.

Mental Health Buddy Support ***(Raspberry Pi, IoT, ChatGPT API)*** — September 2025 – December 2025

- Designed and built an interactive mental health device using a Raspberry Pi and SenseCAP Indicator IoT display, integrating touch, temperature, and light sensors for user control.
- Integrated the ChatGPT API to deliver real-time, conversational mental health support, leveraging stored data for personalized outputs and automatic brightness adjustment based on ambient light levels.
- Developed personalized mental health tools such as daily mood check-in, calming audio, interactive “Buddy” chatbot, and a fidget-style mini-game.

INVOLVEMENT & ACHIEVEMENT

- Launch Your Future Engineering (LYFE) - *Mentor* — May 2025 – Present
- Institute of Electrical and Electronics Engineers (IEEE) - *Active Member* — January 2025 – Present
- Dean’s List — December 2025
- 2 x President’s Honor List — December 2024 – May 2025
- Whitacre College of Engineering Scholarship — August 2025 – Present
- Texas Tech Presidential Scholarship — August 2024 – Present